ME410: Robotics, HW 1
Homework due on 01-19-2025, Topics: Introduction to Python and MuJoCo.

1. **Installing Python IDE:** This question does not have a submission but is needed to answer the questions below. Install any Python IDE. I recommend installing Microsoft Visual Studio Code. Please setup python (e.g., YouTube video https://youtu.be/9o4gDQvVkLU) test a simple program to ensure it works. But feel free to use any other IDE. Just search. e.g., Python IDEs from the web
No Submission for this problem

2. **Python numpy data types:** The questions below consider the matrix and vector below: $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. There are two products/sums we are interested in. $Ab = \begin{bmatrix} 4 \\ 11 \end{bmatrix}$ and $Ab + b = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$ Your goal for each question is to write the correct Python expression for submission. When testing out, dont forget to import numpy as np else you will get an error.

   (a) Define A, b_1, and b_2 as follows:
   A = np.array([[1, 2], [3, 5]])
   b1 = np.array([[2], [1]])
   b2 = np.array([2, 1])
   Write the correct Python expression to compute Ab for two b's given above. e.g., Submission would be simply: $A * b1$ and $A * b2$ and similar for other problems

   (b) Define A, b_1, and b_2 as follows:
   A = np.matrix([[1, 2], [3, 5]])
   b1 = np.array([[2], [1]])
   b2 = np.array([2, 1])
   Write the correct Python expression to compute Ab for two b's given above

   (c) Define A, b_1, and b_2 as follows:
   A = np.array([[1, 2], [3, 5]])
   b1 = np.matrix([[2], [1]])
   b2 = np.array([2, 1])
   Write the correct Python expression to compute $Ab1+b2$ where the b's are defined above

   (d) Define A = np.array([[1, 2], [3, 5]]) and write the expression needed to compute the inverse $\mathbf{A}^{-1}$ and the transpose A^T.

   - **Gradescope:** Submit your Python expression for each problem. No need to show the calculations.
   - **Email:** None

3. **Python functions:** Consider two vectors $\vec{a} = a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k}$ and $\vec{b} = b_x\hat{\imath} + b_y\hat{\jmath} + b_z\hat{k}$. The cross product is $\vec{a} \times \vec{b} = (a_yb_z - a_zb_y)\,\hat{\imath} - (a_xb_z - a_zb_x)\,\hat{\jmath} + (a_xb_y - a_yb_x)\,\hat{k}$. The cross product can also be written using the skew symmetric matrix: $\vec{a} \times \vec{b} = \mathbf{S}(a)\mathbf{b}$ where $\mathbf{S}$ is a 3×3 matrix $\mathbf{S}(\mathbf{a}) = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$. Write a function for skew symmatrix matrix, $\mathbf{S}$,

and evaluate the cross product for $\mathbf{a} = [1, 2, 3]$ and $\mathbf{b} = [4, 5, 6]$ using the function you created and using $np.cross()$. Compare the two.

- **Gradescope:** Submit your Python code and final answer. All submssion in Gradescope are pdf (simply copy paste into word or other editor and output as pdf).
- **Email:** None

4. **Python animations:** Create an animated emoji from the list below (or feel free to search for others) https://googlefonts.github.io/noto-emoji-animation/ of your choice.
   Submission:
   - **Gradescope:** None
   - **Email:** Submit a video or a YouTube link to pranav@uic.edu. Please add "HW1 Animated emoji" to your subject line. If submitting as a group, please copy your group member.

5. **MuJoCo Rube Goldberg Machine:** You aim to create a cascading effect like that in the Rube Goldberg machine using geometries available in MuJoCo. You can also arrange by different orientation/position/size/shape to create more interesting behaviors. Use at least 10 geometries. Here is one example from YouTube. https://youtu.be/OHwDf8njVfo?t=85.
   Learning material: Please watch this video for a tutorial on using XML files:
   https://youtu.be/09pnDvumzvo.
   Install MuJoCo: If you have not installed MuJoCo you can do that by typing pip install mujoco in terminal (mac/Ubuntu)/cmd (windows) or alternately watch this short video:
   https://youtube.com/shorts/1NMBDDLC_gM?feature=share
   Submission:
   - **Gradescope:** None
   - **Email:** Submit the XML file to pranav@uic.edu. Please add "HW1 MuJoCo XML" to your subject line. If submitting as a group, please copy your group member.